



4927 - Testing The Polar Vortex Hypothesis: JWST Predictions for Spatially Distinct Circulation Modes and Composite Spectrum in the Polar Regions of Giant Exoplanets and Brown Dwarfs

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The atmospheres of directly imaged exoplanets and brown dwarfs display a significant diversity: Even for the same temperature, their colors and spectra can differ substantially. Multiple processes have been invoked to explain these variations -- all within the framework of one-dimensional models. However, recently, four lines of evidence emerged that suggests an additional -- and perhaps dominant -- process. We introduce here the Polar Vortex Hypothesis and propose a JWST Theory/Archival Program to assess which JWST observations (existing or future) can be used to test this hypothesis.

Polar Vortex Hypothesis: Giant exoplanets and brown dwarfs - just like Jupiter and Saturn - have composite atmospheres that consist of jet-dominated (belts-and-zones) equatorial regions and vortex-dominated polar regions. These regions have distinct colors and spectra, resulting in

inclination-dependent variations in the colors and spectra of brown dwarfs.

Time-resolved multi-scale spectrophotometry has the potential of disentangling the equatorial and polar components: Equatorial/mid-latitude features will rotate in-and-out of the visible hemisphere, i.e., modulated on the rotational timescale. In contrast, the always-visible polar regions (in inclined brown dwarfs) will evolve over much longer timescales.

We will connect existing atmospheric models (1D radiative transfer and 3D general circulation models) to simulate JWST observations to assess if and how these can disentangle equatorial and polar features and spectral components. Our project will add a fundamental dimension to the understanding of brown dwarf atmospheres and the interpretation of existing and future JWST data.

OBSERVING DESCRIPTION